STOP READING. START BUILDING.

Build Your Own Transformer

From attention by hand to a model you train, serve and take apart

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Everything runs on a **free Colab CPU**.

Swipe »

THE GAP

Reading vs Building

✗ Reading about it

- ✗ One diagram of attention
- ✗ Formulas you nod along to
- ✗ Someone else's trained model
- ✗ Stops at the architecture

✓ Building it

- ✓ Attention derived by hand
- ✓ Tests that catch real bugs
- ✓ Your own model, trained
- ✓ Served, adapted, taken apart

You understand it when **you can break it and fix it.**

WHAT YOU BUILD

Meet Your Base Model

tinyformer-lab: your own configurable transformer

The reference model every lesson trains the same way.

800K
parameters (800,512): small enough to train in minutes

4x4
layers and heads, width 128, context 128

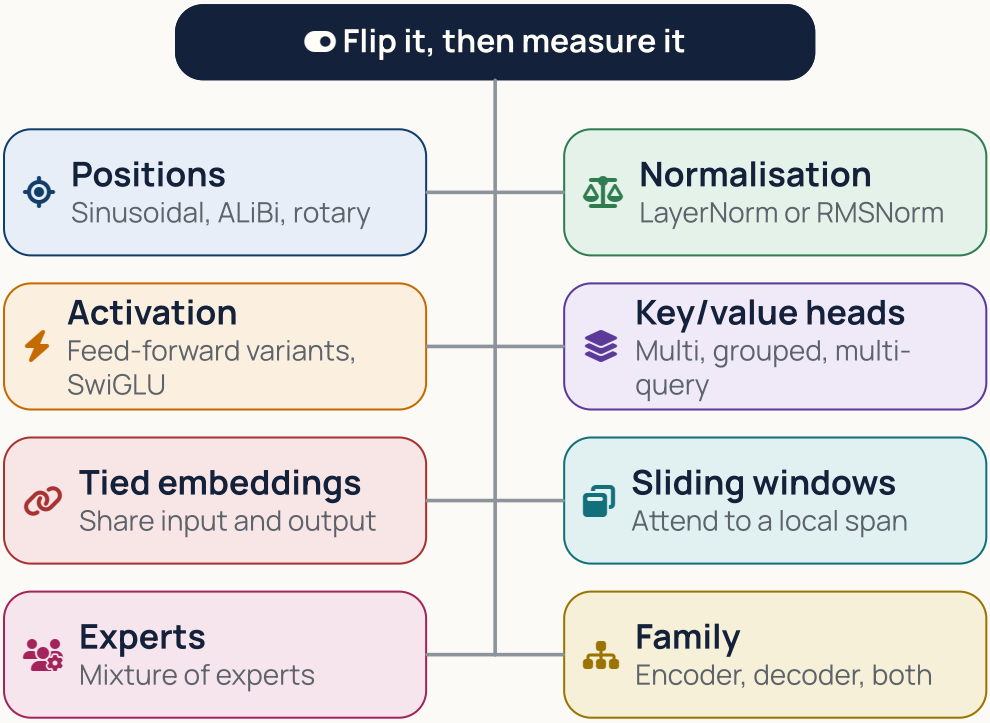
0.256
validation loss after 600 steps

69
tokens: one per character, plus three specials

Small on purpose: **every run finishes on a CPU.**

THE DESIGN

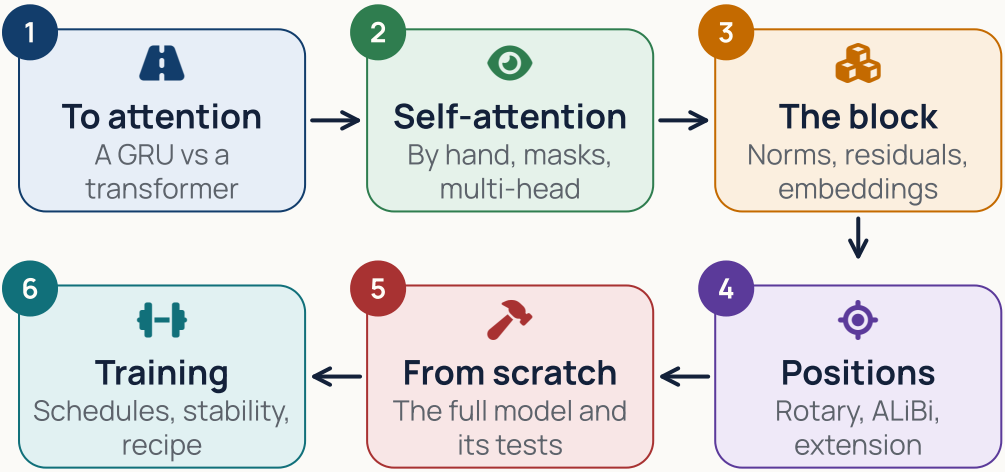
Every Choice Is a Switch



Each switch comes with **a test and a measurement.**

PART 1: BUILD IT

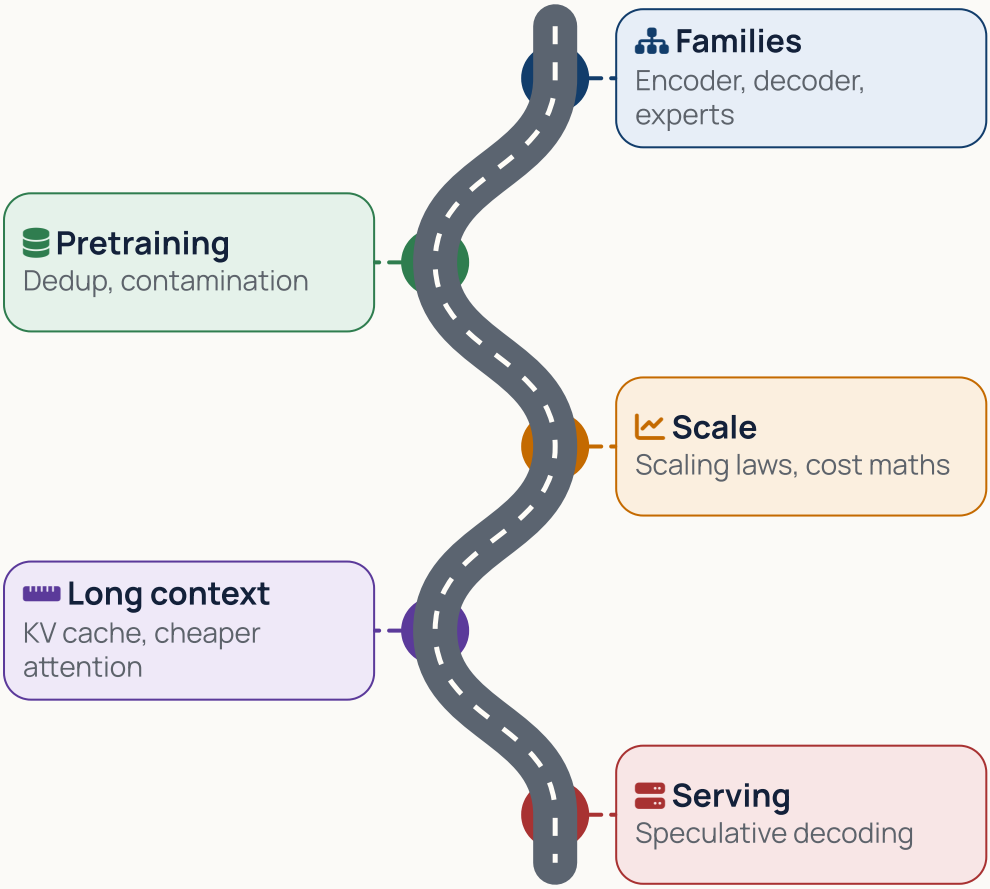
Modules 1 to 6



Plain **PyTorch** first, for every mechanism.

PART 2: SCALE AND SERVE IT

Modules 7 to 11

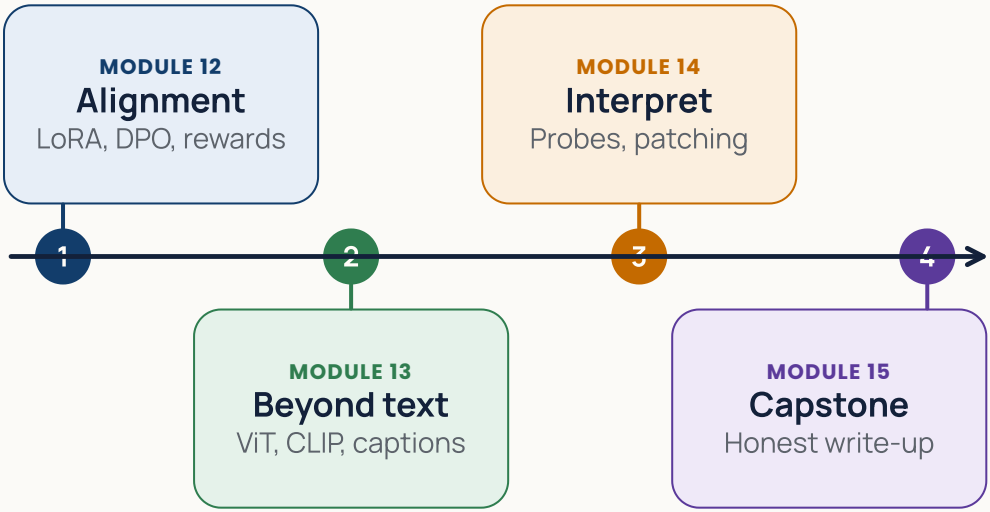


PART 3: ADAPT AND OPEN IT UP

Modules 12 to 15

Then you take your own model apart

And compare what it learnt with GPT-2 small.



The capstone reports **what it cost and what you would not claim.**

THE HOUSE RULE

No Claim Without a Test

Tests that catch real bugs

Every comparison also needs a seed noise floor.

- [x] Causal mask: the model cannot see the future
- [x] Overfit one batch: the model can learn at all
- [x] Gradient check against finite differences
- [x] Same comparison over several seeds
- [x] A run manifest for every result

HONEST NUMBERS

Every Number Is Labelled

Measured
From the lesson's own code, machine named

Cited
From a named source, with a date

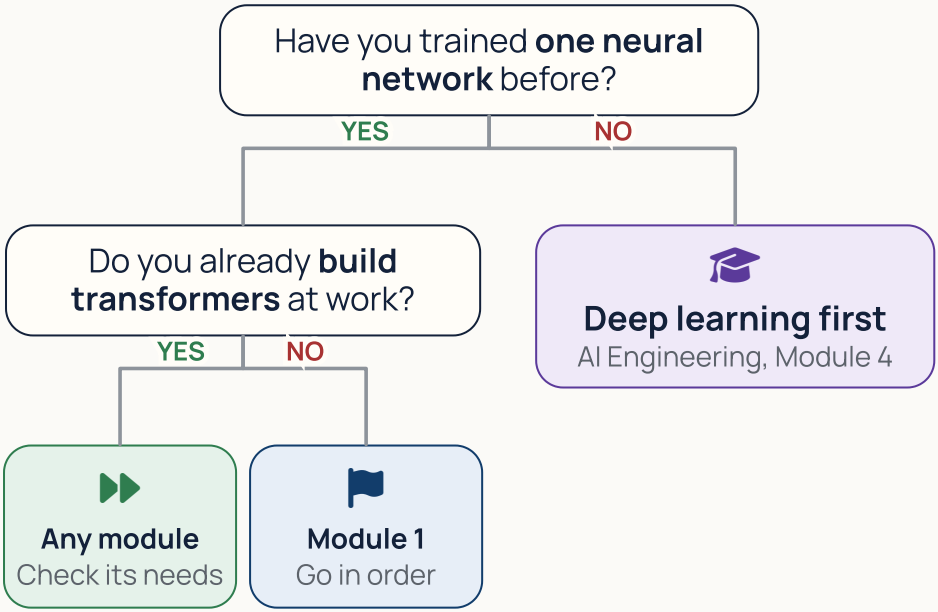
Arithmetic
Worked out from a formula on the page

If a number has no label, it does not go in.

The habit that separates engineers from tutorial readers.

QUICK CHECK

Is This Course for You?



You need Python, NumPy, the **chain rule and matrix products.**

STOP READING ABOUT ATTENTION

Build, Train and Open Up a Transformer

One repository, every design choice a switch you flip and measure

- 14 modules plus a capstone, attention to interpretability
- Runs on the free Colab CPU runtime, no GPU
- Plain PyTorch first, then Hugging Face and peft
- A test for every claim the lab makes

Start the Transformers course ↓

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Questions? **Comment below** and I'll answer every one.